EXERCISE 10.7 DISCRETE RANDOM VARIABLES

1 Information about the numbers from 1 to 50 inclusive is contained in the following table.

	Multiple of 3	Multiple of 7	Total
Odd	8	4	12
Even	8	3	11
Total	16	7	23

(a) How many multiples of 3 are there?
(b) Which two numbers have been included in two different groups?
(c) How many numbers from 1 to 50 inclusive are not included in this table?

2 In a survey, of 100 high school students were asked what type of phone they owned. The results are contained in the following table.

	Apple	Samsung	LG	Other	Total
Boys	15	12	15	8	50
Girls	20	10	15	5	50
Total	35	22	30	13	100

(a) How many students had an Apple phone?
(b) How many students did not have a Samsung phone?
(c) What proportion of the students had a LG phone?
(d) If a student was selected at random from the group, what is the probability that their phone was not an Apple, Samsung or LG?

3 **(a)** Copy the given table and add to it a cumulative frequency column. What is the modal class?

Score	Frequency
0–<10	9
10–<20	18
20–<30	13
30–<40	15
40–<50	14
50–<60	16

Draw the ogive for this table and use it to find the median.

(b) Copy the given table and add to it a cumulative frequency column.

Score	Frequency
141–<146	22
146–<151	31
151–<156	21
156–<161	26
161–<166	22
166–<171	24

What is the modal class? Draw the ogive for this table and use it to find the median.

4 A survey that asked students their favourite sport produced the information in the following table.

	Football	Swimming	Basketball	Total
Male	32	30		70
Female		28	12	
Total	50		20	

(a) Copy the table and fill in the missing information.
(b) How many students gave swimming as their favourite sport?
(c) What proportion of the males gave basketball as their favourite sport?
(d) A student is chosen at random from the group. What is the probability that they gave football as their favourite sport?

5 The following data represents the number of kilometres travelled each day by a solar vehicle. The distances have been rounded to the nearest whole number of kilometres.

112, 95, 122, 78, 99, 145, 133, 160, 145, 144, 78, 66, 68, 79, 93, 125, 133, 89, 67, 78, 114, 134, 80, 65, 105, 115, 127, 134, 117, 84

(a) Given the way the data has been recorded, would you consider it to be discrete or continuous? Give a brief justification for your answer.
(b) Construct a frequency table for the data, using a class interval of 10 and a starting value of 61.
(c) Looking only at the frequency table, what can you say about the maximum distance travelled on any one day?

6 A manufacturer received the following complaints from purchasers of their new ovens. Add a relative frequency column to the table.

Defect	Frequency
Faulty light	17
Faulty fan	12
Faulty element	9
Damaged box	4
Scratched glass	3
Other	2
Total	**47**

(a) What is the probability of the complaint being a faulty element?
(b) What is the probability of the complaint being a faulty light?

7 A hotel chain conducted a survey of their guests and received the following responses. Add a relative frequency column to the table.

Subject of complaint	Frequency
Housekeeping	45
Check-in/checkout	25
Breakfast	15
Value for money	9
WiFi	6
Television	4
Parking	2
Total	**106**

(a) What is the probability of the complaint being about breakfast?
(b) What complaint has a more than 40% chance of being received by the hotel?

8 Find the mode and the median for each set of scores.

(a) 1, 4, 3, 2, 6, 4, 8, 9, 4, 3, 1
(b) 2, 3, 6, 2, 9, 6, 9, 6, 3, 2, 5, 2
(c)

Score (x)	Frequency (f)
0	3
1	6
2	7
3	4
4	3
5	1

9 Find the mode and the median for each set of scores. Where necessary, state your answers correct to 2 decimal places.

(a) The following data represents the number of cars that passed the school gate in a number of 5-minute periods.
7, 14, 12, 11, 13, 16, 17, 9, 7, 12, 13, 15
(b) The following data represents the number of brothers and sisters for the students in two Year 11 classes.

Number of brothers and sisters (x)	Frequency (f)
0	6
1	11
2	18
3	9
4	3

(c) The following data represents the time taken by the competitors in the 17-and-under 100 m dash at the school athletics carnival.

Time (x)	Frequency (f)
12–<12.5	4
12.5–<13	9
13–<13.5	15
13.5–<14	22
14–<14.5	17

10 Find the mode of the following data sets.

(a) 4, 6, 4, 2, 7, 8, 3, 4, 9, 1, 1 **(b)** 3, 5, 2, 1, 7, 4, 9, 3, 5, 6, 8 **(c)** 1, 4, 2, 1, 5, 4, 3, 1, 4, 2, 2

11 Find the median class interval for the following data sets.

(a)

Time (x)	Frequency (f)
12–<12.5	3
12.5–<13	5
13–<13.5	4
13.5–<14	11
14–<14.5	6

(b)

Time (x)	Frequency (f)
12–<12.5	6
12.5–<13	8
13–<13.5	4
13.5–<14	5
14–<14.5	7

12 Consider the following data set.

x	1	2	3	4	5	6
f	5	8	3	2	4	5

The mode and median (in that order) are:

A 2, 3 **B** 3.26, 3 **C** 3.26, 4 **D** 3, 3.26

13 Find the median for each of the following discrete data sets. Where necessary, state your answer correct to 2 decimal places.

(a)

Score	Frequency (f)
4	15
5	23
6	14
7	23
8	17
9	19
10	20

(b)

Score	Frequency (f)
201	49
202	83
203	99
204	121
205	143
206	117

14 The following table gives the maximum temperature (°C) reached on a particular day in a number of cities across the world. Answer the following questions about this data set.

City	Temp (°C)	City	Temp (°C)	City	Temp (°C)
Amsterdam	3	Copenhagen	0	New York	13
Athens	18	Helsinki	–4	Oslo	–3
Barcelona	9	Istanbul	18	Prague	0
Belgrade	3	London	5	Singapore	33
Budapest	1	Los Angeles	14	Taipei	27
Cairo	34	Madrid	6	Toronto	10
Chicago	13	Moscow	1	Warsaw	–1

(a) Find the mode of the temperatures.
(b) Calculate the median temperature.
(c) Draw a grouped data frequency table using –5 to <0 as the first class interval.
(d) In which class interval is the median temperature?

15 Consider the following frequency tables:

(a)

Score	Frequency
44	8
45	12
46	13
47	10
48	9
49	5

(b)

Score	Frequency
101	11
102	14
103	18
104	19
105	13
106	12

(i) Copy the table and add a relative frequency column.
(ii) What is the median for the data?
(iii) What is the probability that the score will be the:
(α) smallest one in the table? (β) largest one in the table?

16 Consider the following frequency tables:

(a)

Score	Frequency
10–14	6
15–19	7
20–24	9
25–19	10
30–34	4
35–39	1

(b)

Score	Frequency
80–<90	16
90–<100	15
100–<110	17
110–<120	19
120–<130	12
130–<140	11

(i) Copy the table and add a cumulative frequency column.
(ii) Draw the cumulative frequency polygon for the data and use it to estimate the median (class).
(iii) What is the probability that the score will be in the:
(α) smallest class interval (β) largest class interval?

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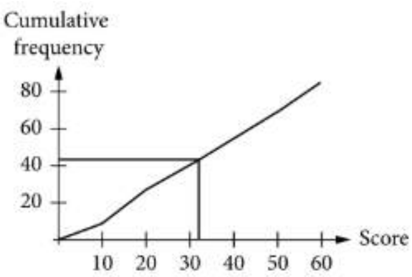
1 (a) 16 (b) 21 and 42 (c) 29

2 (a) 35 (b) 78 (c) 30% (d) 0.13

3 (a) Modal class = 10–<20

Score	Frequency	Cumulative frequency
0–<10	9	9
10–<20	18	27
20–<30	13	40
30–<40	15	55
40–<50	14	69
50–<60	16	85

Median is the 43rd score.

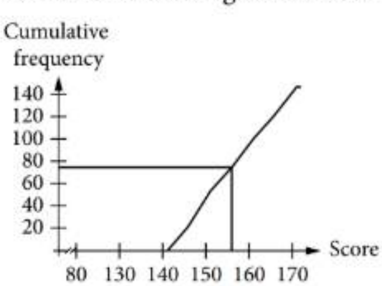


Median is 32.

(b) Modal class = 146–<151

Score	Frequency	Cumulative frequency
141–<146	22	22
146–<151	31	53
151–<156	21	74
156–<161	26	100
161–<166	22	122
166–<171	24	146

Median is the average of the 73rd and 74th scores.



Median is 156.

4 (a)

	Football	Swimming	Basketball	Total
Male	32	30	8	70
Female	18	28	12	58
Total	50	58	20	128

(b) 58 (c) $\frac{4}{35}$ (d) $\frac{25}{64}$

5 (a) Data is continuous because it has been measured.

(b)

Kilometres travelled	Frequency
61–<71	4
71–<81	5
81–<91	2
91–<1019	3
101–<111	1
111–<121	4
121–<131	3
131–<141	4
141–<151	3
151–<161	1

(c) Maximum distance travelled is between 151 and 161 km.

(d)

Stem	Leaf
6	5 6 7 8
7	8 8 8 9
8	9 0 4
9	3 5 9
10	5
11	2 4 5 7
12	2 5 7
13	3 3 4 4
14	4 5 5
15	
16	0

Key: 6|5 = 65

(e) The stem plot gives the actual data values. The frequency table with grouped data does not show individual data values.

(f) Stem plots show actual discrete data values, whereas frequency tables do not.

6 (a)

Defect	Frequency	Relative frequency
Faulty light	17	$\frac{17}{47} \approx 0.362$
Faulty fan	12	$\frac{12}{47} \approx 0.255$
Faulty element	9	$\frac{9}{47} \approx 0.191$
Damaged box	4	$\frac{4}{47} \approx 0.085$
Scratched glass	3	$\frac{3}{47} \approx 0.064$
Other	2	$\frac{2}{47} \approx 0.043$
Total	47	1

(a) $\frac{9}{47} \approx 0.191$

(b) $\frac{17}{47} \approx 0.362$

7 (a)

Subject of complaint	Frequency	Cumulative percentage
Housekeeping	45	$\frac{45}{106} \approx 0.425$
Check-in/ checkout	25	$\frac{25}{106} \approx 0.234$
Breakfast	15	$\frac{15}{106} \approx 0.142$
Value for money	9	$\frac{9}{106} \approx 0.085$
WiFi	6	$\frac{6}{106} \approx 0.057$
Television	4	$\frac{4}{106} \approx 0.038$
Parking	2	$\frac{2}{106} \approx 0.019$
Total	**106**	1

(a) $\frac{15}{106} \approx 0.142$

(b) Poor housekeeping

8 (a) Mode = 4, Median = 4
(b) Mode = 2, Median = 4
(c) Mode = 2, Median = 2

9 (a) Mode = 7, 12, 13, Median =12.5
(b) Mode = 2, Median = 2
(c) Modal class = 13.5–<14, Median = 13.6 (13.5–<14)

10 (a) 4 (b) 3 and 5 (c) No mode

11 (a) The median is the interval 13.5–<14
(b) The median is the interval 13–<13.5

12 A; Mode = 2, Median = 3

13 (a) median is 7
(b) median is 204

14 (a) 0 and 1°C (b) 6°C

(c)

Temperature range	Frequency
−5–<0	3
0–<5	6
5–<10	3
10–<15	4
15–<20	2
20–<25	0
25–<30	1
30–<35	2

(d) 5–<10

15 (a) (i)

Score	Frequency	Relative frequency
44	8	$\frac{8}{57} = 0.140$
45	12	$\frac{12}{57} = 0.211$
46	13	$\frac{13}{57} = 0.228$
47	10	$\frac{10}{57} = 0.175$
48	9	$\frac{9}{57} = 0.158$
49	5	$\frac{5}{57} = 0.088$
Total	57	1

(ii) 46 (iii) (α) $\frac{8}{57} = 0.140$ (β) $\frac{5}{57} = 0.088$

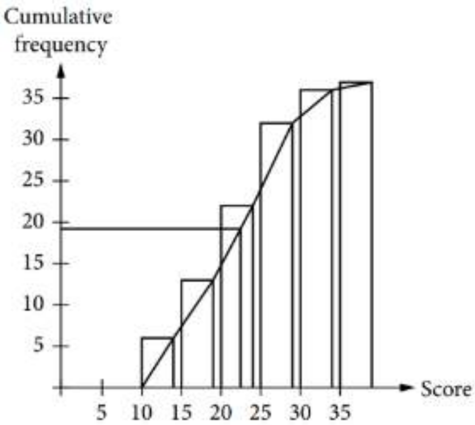
(b) (i)

Score	Frequency	Relative frequency
101	11	$\frac{11}{87} = 0.126$
102	14	$\frac{14}{87} = 0.161$
103	18	$\frac{18}{87} = 0.207$
104	19	$\frac{19}{87} = 0.218$
105	13	$\frac{13}{87} = 0.149$
106	12	$\frac{12}{87} = 0.138$
Total	87	1

(ii) 104 (iii) (α) $\frac{11}{87} = 0.126$ (β) $\frac{12}{87} = 0.138$

16 (a) (i)

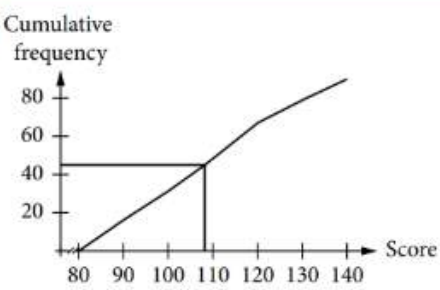
Score	Frequency	Cumulative frequency
10–14	6	6
15–19	7	13
20–24	9	22
25–29	10	32
30–34	4	36
35–39	1	37
Total	37	

(ii) 

(iii) (α) $\frac{6}{37} = 0.162$ (β) $\frac{1}{37} = 0.027$

(b) (i)

Score	Frequency	Cumulative frequency
80–<90	16	16
90–<100	15	31
100–<110	17	48
110–<120	19	67
120–<130	12	79
130–<140	11	90
Total	90	

(ii) 

(iii) (α) $\frac{16}{90} = 0.178$ (β) $\frac{11}{90} = 0.122$